Smooth Graph Representations of Travel Time on SE(3)

A complete mathematical derivation of the graph–decoder idea

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Abstract

The minimum travel time $T(x, y)$ between two rigid-body poses is a non-smooth function: it has a cone-shaped cusp where $x = y$ and a corner where the relative rotation is a half turn. This report develops, with complete derivations, the idea of representing T through a *smooth* parametrization H of a transformed travel-time graph followed by an explicit, possibly non-smooth, *decoder* D, so that $T = D \circ H$, and of imposing the Eikonal equation through the chain rule $\mathrm{d}T = \mathrm{d}D \circ \mathrm{d}H$. Part I states the idea on a general configuration manifold. Part II specializes it to SE(3): we set up the Lie-group geometry from scratch, derive the Eikonal equation from the motion model, compute the exact travel time for an ellipsoidal speed limit, exhibit an explicit smooth two-scalar latent $Z = (a, b)$ with an explicit decoder, verify the Eikonal equation through H, identify a latent Riemannian metric in which the decoder is a distance function, analyse the singular sets, and formulate the learning problem (Eikonal, boundary and Bellman losses). Every claim is either proved here or explicitly marked as a hypothesis.

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Notation at a glance

Symbol	Meaning
$x = (t_x, R_x) \in \mathrm{SE}(3)$	pose: translation $t_x \in \mathbb{R}^3$, rotation $R_x \in \mathrm{SO}(3)$
$p = (x, y) \in N = \mathrm{SE}(3) \times \mathrm{SE}(3)$	a (start, goal) pose pair; $\dim N = 12$
$[\omega]_\times$	skew matrix with $[\omega]_\times u = \omega \times u$
$(\cdot)^\vee$	inverse of $[\cdot]_\times$: $([\omega]_\times)^\vee = \omega$
$\xi = (v, \omega) \in \mathbb{R}^6$	body twist (linear, angular velocity in the body frame)
$s_t, s_r > 0$	translational / rotational speed scales
$\xi_1, \dots, \xi_6$	scaled twist frame: $\xi_i = (s_t e_i, 0)$, $\xi_{3+j} = (0, s_r e_j)$
$c_x, c_y \in \mathbb{R}^6$	endpoint gradients of T in the scaled frame
$G(x) = (t_x, \operatorname{vec} R_x) \in \mathbb{R}^{12}$	fixed pose embedding
$Z(p) = (a, b) \in \mathbb{R}^2$	smooth latent ("transformed time")
$H(p) = (G(x), G(y), Z(p)) \in \mathbb{R}^{26}$	graph parametrization; image $\Sigma = H(N)$
$D(a, b)$	decoder, $T = D \circ H$
$\theta(R) \in [0, \pi]$	rotation angle; $\theta(x, y) = \theta(R_x^\top R_y)$
$\varphi(b) = 4 \arcsin^2 \sqrt{b}$	rotational part of the decoder

Vectors are columns; $|\cdot|$ is the Euclidean norm on $\mathbb{R}^n$, $\|\cdot\|_F$ the Frobenius norm of a matrix, and e_1, e_2, e_3 the standard basis of $\mathbb{R}^3$.

Part I
The general idea

1 Graph, transformation and decoder

1.1 Setting

Let M be a configuration space of dimension d (later $M = \mathrm{SE}(3)$, $d = 6$), and let $N = M \times M$ be the space of pairs $p = (x, y)$ (start x, goal y), of dimension $2d$. Let

$$T : N \to \mathbb{R}_{\geq 0}$$

be the minimum travel time from x to y. We assume T is finite and continuous. The *graph* of T is

$$\Gamma_T = \{(p, T(p)) : p \in N\} \subset N \times \mathbb{R}. \tag{1.1}$$

Because T is continuous, the projection $(p, T(p)) \mapsto p$ is a homeomorphism $\Gamma_T \cong N$. If T were smooth, Γ_T would be a smooth manifold; it is not, because T has cusps and corners (Section 8).

1.2 Transforming the graph

The idea is to replace Γ_T by a transformed copy that *is* smooth. A transformation of the whole graph is a map

$$\Phi\big(p, T(p)\big) = H(p), \qquad H : N \to \mathbb{R}^m. \tag{1.2}$$

We use the structured form that keeps the configurations and transforms only the time information:

$$H(p) = \big(B(p), Z(p)\big), \qquad B(x,y) = \big(G(x), G(y)\big), \qquad Z(p) = F\big(p, T(p)\big) \in \mathbb{R}^k. \tag{1.3}$$

Here $G : M \to \mathbb{R}^n$ is a fixed smooth embedding of M, and F may depend on all of p and on the time; in particular Z need not be a function of T alone, and k may be larger than 1. The image

$$\Sigma = H(N) \subset \mathbb{R}^{2n+k} \tag{1.4}$$

is the *transformed graph*.

Proposition 1.1 (Embedding)**.** *If G is a smooth embedding and Z is smooth, then $H = (B, Z)$ is a smooth embedding and Σ is a smooth $2d$-dimensional submanifold of $\mathbb{R}^{2n+k}$.*

Proof. Injective: if $H(p) = H(p')$ then $G(x) = G(x')$ and $G(y) = G(y')$, so $p = p'$. *Immersion:* for a tangent vector w at p, $\mathrm{d}H_p(w) = (\mathrm{d}B_p(w), \mathrm{d}Z_p(w))$; if this is zero then $\mathrm{d}B_p(w) = 0$, and $\mathrm{d}B$ is injective since $\mathrm{d}G$ is, so $w = 0$. *Homeomorphism onto the image:* the inverse $\Sigma \to N$ is the restriction of the continuous map $(b, z) \mapsto B^{-1}(b)$, which is continuous because B is an embedding. □

Remark 1.2. Keeping B fixed is what makes Proposition 1.1 automatic. If every coordinate of H were the output of an unconstrained neural network, H need not be injective nor have full rank.

1.3 The decoder

A *decoder* is a function

$$D : \Sigma \to \mathbb{R}_{\geq 0} \qquad \text{with} \qquad T = D \circ H. \tag{1.5}$$

Since B recovers p, the inverse of H on Σ recovers p and D only has to recover time; it is the time component of H^{-1}, not the whole inverse. In all examples below D depends only on the Z-coordinates.

Proposition 1.3 (Where the non-smoothness must go)**.** *If H is smooth and D is smooth on a neighbourhood of Σ, then $T = D \circ H$ is smooth. Consequently, if T fails to be differentiable at p_0 and H is smooth, then D must fail to be differentiable at $H(p_0)$.*

Proof. Composition of smooth maps is smooth. □

So the idea is not to make T smooth (impossible) but to move all of its non-smoothness into an *explicit, known* decoder, leaving a smooth object H (in practice, a smooth Z) to be learned.

2 Pulling the Eikonal equation through H

2.1 The Eikonal equation with an input metric

Equip N with an inner product g_p on each tangent space (a Riemannian metric). For a differentiable function f, write $\|\mathrm{d}f_p\|_{g^{-1}}$ for the dual norm

$$\|\mathrm{d}f_p\|_{g^{-1}} = \sup_{g_p(w,w)=1} |\mathrm{d}f_p(w)|. \tag{2.1}$$

If $w_1, \ldots, w_{2d}$ is a g_p-orthonormal basis, then $\|\mathrm{d}f_p\|^2_{g^{-1}} = \sum_i \mathrm{d}f_p(w_i)^2$, i.e. the ordinary squared norm of the vector of directional derivatives. The (joint) Eikonal equation with speed $S > 0$ is

$$\|\mathrm{d}T_p\|^2_{g^{-1}} = S(p)^{-2}. \tag{2.2}$$

2.2 Chain rule

At a point where T, H and D (restricted to Σ) are differentiable, for every tangent vector w,

$$\mathrm{d}T_p(w) = \mathrm{d}D_{H(p)}\big(\mathrm{d}H_p(w)\big). \tag{2.3}$$

The vector $\mathrm{d}H_p(w)$ is tangent to Σ; $\mathrm{d}D$ measures the change of decoded time along it. In coordinates: suppose D extends to a differentiable function on a neighbourhood of Σ in $\mathbb{R}^m$ (still denoted D), let $J_H(p) \in \mathbb{R}^{m\times 2d}$ be the Jacobian of H in a g-orthonormal frame and $w = \nabla D(H(p)) \in \mathbb{R}^m$. Then

$$\nabla_p T = J_H(p)^\top \nabla D(H(p)), \tag{2.4}$$

and the Eikonal equation becomes a constraint on the parametrization:

$$\nabla D(H(p))^\top J_H(p) J_H(p)^\top \nabla D(H(p)) = S(p)^{-2}. \tag{2.5}$$

Proposition 2.1 (Independence of the ambient extension)**.** *Let D_1, D_2 be two differentiable extensions of D that agree on Σ. Then $J_H^\top \nabla D_1 = J_H^\top \nabla D_2$ at every point of Σ.*

Proof. Let u be tangent to Σ at q and $\gamma(s)$ a curve in Σ with $\gamma(0) = q$, $\gamma'(0) = u$. Since $D_1 \circ \gamma = D_2 \circ \gamma$, differentiating at $s = 0$ gives $\nabla D_1(q) \cdot u = \nabla D_2(q) \cdot u$. Thus $\nabla D_1 - \nabla D_2$ is orthogonal to $T_q\Sigma$. The columns of J_H span $T_q\Sigma$, so $J_H^\top(\nabla D_1 - \nabla D_2) = 0$. □

In practice no extension is needed: one simply differentiates the scalar composite $p \mapsto D(H(p))$. Equation (2.5) is therefore an *equivalent* re-expression of (2.2) on the set where everything is differentiable; introducing the graph does not create a new or stronger PDE.

2.3 What is constrained geometrically

With $h_i = \mathrm{d}H_p(w_i)$ for a g-orthonormal basis (w_i),

$$\sum_{i=1}^{2d} \big[\mathrm{d}D_{H(p)}(h_i)\big]^2 = S(p)^{-2}, \qquad \text{equivalently} \qquad \sup_{g(w,w)=1} |\mathrm{d}D_{H(p)}(\mathrm{d}H_p w)| = S(p)^{-1}. \tag{2.6}$$

The PDE fixes the *largest decoded time change per unit input motion.* It does not by itself fix which direction realises it; boundary conditions and global consistency do.

Transported metric. Define a metric $\bar{g}$ on Σ by declaring $\mathrm{d}H_p$ an isometry:

$$\bar{g}_{H(p)}\big(\mathrm{d}H_p v, \mathrm{d}H_p u\big) = g_p(v, u). \tag{2.7}$$

Then, for $q \in \Sigma$,

$$\|\mathrm{d}D_q\|^2_{\bar{g}^{-1}} = S\big(H^{-1}(q)\big)^{-2}. \tag{2.8}$$

Proof: $\sup_{\bar{g}(u,u)=1} |\mathrm{d}D_q(u)| = \sup_{g(v,v)=1} |\mathrm{d}D_q(\mathrm{d}H_p v)| = \|\mathrm{d}T_p\|_{g^{-1}}$.

Not the ambient induced metric. The Euclidean metric of $\mathbb{R}^m$ restricted to Σ gives, in the coordinates p, the matrix $h = J_H^\top J_H$, and the corresponding squared gradient norm of $D|_\Sigma$ is $w^\top J_H (J_H^\top J_H)^{-1} J_H^\top w$. This equals the left side of (2.5) only when $J_H^\top J_H = I$, i.e. when H is an isometric embedding. That is *not* assumed, and for SE(3) it is false (Remark 7.8). Hence the loss constrains the triple (parametrization, metric, decoder) together; it does not prescribe the curvature or Euclidean "tilt" of Σ by itself.

Part II
Specialization to SE(3)

3 Geometry of rotations and rigid motions

This section collects everything about SO(3) and SE(3) that is used later. No smooth manifold theory is assumed: tangent vectors are introduced as velocities of curves of matrices.

3.1 Skew matrices and their identities

For $\omega = (\omega_1, \omega_2, \omega_3) \in \mathbb{R}^3$ define

$$[\omega]_\times = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}, \qquad [\omega]_\times u = \omega \times u. \tag{3.1}$$

Its inverse is $(\cdot)^\vee$: $([\omega]_\times)^\vee = \omega$. The set of 3×3 skew matrices is $\mathfrak{so}(3)$.

Lemma 3.1. *For $\omega, \mu \in \mathbb{R}^3$ and $R \in \mathrm{SO}(3)$:*

(a) $[\omega]_\times [\mu]_\times = \mu\omega^\top - (\omega \cdot \mu) I$;

(b) $\mathrm{tr}([\omega]_\times [\mu]_\times) = -2\,\omega \cdot \mu$, *and* $\|[\omega]_\times\|_F^2 = 2|\omega|^2$;

(c) for unit n: $[n]_\times^2 = nn^\top - I$ *and* $[n]_\times^3 = -[n]_\times$;

(d) $R[\omega]_\times R^\top = [R\omega]_\times$;

(e) for any $A \in \mathbb{R}^{3\times 3}$: $\mathrm{tr}([\omega]_\times A) = -\,\omega \cdot (A - A^\top)^\vee$.

Proof. (a) Apply both sides to u: $\omega \times (\mu \times u) = \mu(\omega \cdot u) - u(\omega \cdot \mu)$ (the BAC–CAB rule). (b) Take the trace of (a): $\omega \cdot \mu - 3\,\omega \cdot \mu$. Also $\|[\omega]_\times\|_F^2 = \mathrm{tr}([\omega]_\times^\top [\omega]_\times) = -\,\mathrm{tr}([\omega]_\times^2) = 2|\omega|^2$. (c) Put $\omega = \mu = n$ in (a), then multiply by $[n]_\times$ using $[n]_\times n = 0$. (d) $R[\omega]_\times R^\top u = R(\omega \times R^\top u) = (R\omega) \times u$ since rotations preserve cross products. (e) Write $A = A_s + A_k$ with A_s symmetric and $A_k = \frac{1}{2}(A - A^\top)$ skew. $\mathrm{tr}([\omega]_\times A_s) = 0$ (trace of skew times symmetric), and by (b) $\mathrm{tr}([\omega]_\times A_k) = -2\,\omega \cdot A_k^\vee = -\omega \cdot (A - A^\top)^\vee$. □

3.2 The rotation group and its exponential

$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^\top R = I,\ \det R = 1\}$. If $R(s)$ is a differentiable curve in SO(3), differentiating $R^\top R = I$ shows $R^\top \dot{R}$ is skew, so we can write

$$\dot{R} = R\,[\omega]_\times, \qquad \omega \in \mathbb{R}^3, \tag{3.2}$$

and call ω the *body angular velocity*. The exponential map is the matrix exponential $\exp([\omega]_\times) = \sum_{k\geq 0} [\omega]_\times^k / k!$.

Lemma 3.2 (Rodrigues)**.** *Write $\omega = \theta n$ with $\theta = |\omega|$, $|n| = 1$. Then*

$$\exp(\theta[n]_\times) = I + \sin\theta\,[n]_\times + (1-\cos\theta)[n]_\times^2, \tag{3.3}$$

which is the rotation by angle θ about axis n. Moreover

$$\operatorname{tr}\exp(\theta[n]_\times) = 1 + 2\cos\theta, \qquad \exp(\theta[n]_\times) - \exp(\theta[n]_\times)^\top = 2\sin\theta\,[n]_\times. \tag{3.4}$$

Proof. By Lemma 3.1(c) the powers cycle: $[n]_\times^{2k+1} = (-1)^k[n]_\times$ and $[n]_\times^{2k+2} = (-1)^k[n]_\times^2$. Grouping the series gives $I + \big(\theta - \frac{\theta^3}{3!} + \cdots\big)[n]_\times + \big(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \cdots\big)[n]_\times^2$. For the trace use $\operatorname{tr}[n]_\times = 0$ and $\operatorname{tr}[n]_\times^2 = -2$. The skew part comes only from the $[n]_\times$ term since $[n]_\times^2$ is symmetric. □

Definition 3.3 (Rotation angle and axis)**.** Every $R \in \mathrm{SO}(3)$ equals $\exp(\theta[n]_\times)$ for some $\theta \in [0,\pi]$ and unit n. The angle is unique and, by (3.4),

$$\theta(R) = \arccos\frac{\operatorname{tr}R - 1}{2} = \operatorname{atan2}\Big(\tfrac{1}{2}|(R - R^\top)^\vee|,\ \tfrac{\operatorname{tr}R-1}{2}\Big) \in [0,\pi]. \tag{3.5}$$

For $0 < \theta < \pi$ the axis is unique: $n = (R - R^\top)^\vee/(2\sin\theta)$. For $\theta = 0$ any axis works; for $\theta = \pi$ exactly two axes $\pm n$ work. The principal logarithm is $\operatorname{Log} R = \theta[n]_\times$ for $\theta < \pi$.

The atan2 form is numerically preferable near $\theta = \pi$, where arccos has infinite slope.

3.3 Unit quaternions

A unit quaternion $q = (q_0, q_v) \in \mathbb{R}\times\mathbb{R}^3$, $q_0^2 + |q_v|^2 = 1$, i.e. a point of the sphere $S^3 \subset \mathbb{R}^4$, defines a rotation $R(q)$ with $R(q)u = q\,(0,u)\,\bar q$ (quaternion products). The facts we need:

(Q1) $R(q)R(q') = R(qq')$, $R(\bar q) = R(q)^\top$ and $R(-q) = R(q)$; every rotation has exactly two preimages $\pm q$ (double cover).

(Q2) The rotation by θ about n is $R(q)$ with $q = \big(\cos\frac{\theta}{2},\ \sin\frac{\theta}{2}\,n\big)$. Hence for $\theta(R) \in [0,\pi]$: $|q_0| = \cos\frac{\theta}{2}$ and $|q_v| = \sin\frac{\theta}{2}$.

(Q3) $\operatorname{tr}R(q) = 4q_0^2 - 1$ (from (Q2) and (3.4): $1 + 2\cos\theta = 4\cos^2\frac{\theta}{2} - 1$).

(Q4) If $q(s)$ is a differentiable curve in S^3 and $R(s) = R(q(s))$ satisfies $\dot R = R[\omega]_\times$, then $\dot q = \frac{1}{2}\,q\,(0,\omega)$ and so $|\dot q| = \frac{1}{2}|\omega|$ (because $|qp| = |q||p|$).

(Q5) For unit p, q: $p\cdot q = \operatorname{Re}(\bar p q)$, the scalar part of the quaternion representing $R(p)^\top R(q)$.

(Q6) The length of any curve on S^3 from p to q is at least the great-circle distance $\arccos(p\cdot q)$.

(Q7) Any continuous curve $R(s)$ in $\mathrm{SO}(3)$ lifts to a continuous curve $q(s)$ in S^3 with $R(s) = R(q(s))$ (path lifting for the double cover).

3.4 The rigid-motion group

A pose is $x = (t_x, R_x) \in \mathrm{SE}(3) = \mathbb{R}^3 \times \mathrm{SO}(3)$, i.e. the matrix $\left(\begin{smallmatrix} R_x & t_x \\ 0 & 1 \end{smallmatrix}\right)$. Matrix multiplication gives

$$x\,y = (t_x + R_x t_y,\ R_x R_y), \qquad x^{-1} = (-R_x^\top t_x,\ R_x^\top), \qquad x^{-1}y = \big(R_x^\top(t_y - t_x),\ R_x^\top R_y\big). \tag{3.6}$$

The pose $x^{-1}y$ is the goal expressed in the frame of the start (the *relative pose*).

Twists and the exponential. A body twist is $\xi = (v, \omega) \in \mathbb{R}^6$ with

$$\xi^\wedge = \begin{pmatrix} [\omega]_\times & v \\ 0 & 0 \end{pmatrix} \in \mathfrak{se}(3), \qquad \exp(\xi^\wedge) = \big(V(\omega)v,\ \exp[\omega]_\times\big), \tag{3.7}$$

$$V(\omega) = \sum_{k\geq 0} \frac{[\omega]_\times^k}{(k+1)!} = I + \frac{1-\cos\theta}{\theta^2}[\omega]_\times + \frac{\theta - \sin\theta}{\theta^3}[\omega]_\times^2, \qquad \theta = |\omega|. \tag{3.8}$$

(The block form follows by computing the powers of $\xi^\wedge$; the closed form uses Lemma 3.1(c) as in Rodrigues.) Near $\theta = 0$ one must use the series $V = I + \frac{1}{2}[\omega]_\times + \frac{1}{6}[\omega]_\times^2 + O(\theta^3)$; the closed form is $0/0$ there.

Lemma 3.4 (Spectrum of V)**.** *$V(\omega)$ fixes the axis direction ($V\omega = \omega$) and acts on the plane $\omega^\perp$ as multiplication by the complex number $(e^{i\theta} - 1)/(i\theta)$. In particular $|Vu| = |u|$ for $u \parallel \omega$ and $|Vu| = \frac{\sin(\theta/2)}{\theta/2}|u|$ for $u \perp \omega$, so for every u*

$$|V(\omega)^{-1}u| \leq \kappa(\theta)|u|, \qquad \kappa(\theta) = \frac{\theta/2}{\sin(\theta/2)} \quad (0 \leq \theta < 2\pi). \tag{3.9}$$

Proof. On $\omega^\perp$, $[\omega]_\times$ acts as rotation by 90° scaled by θ, i.e. as multiplication by $i\theta$. So V acts as $\sum_k (i\theta)^k/(k+1)! = (e^{i\theta} - 1)/(i\theta)$, whose modulus is $|2\sin(\theta/2)|/\theta$. On the axis $[\omega]_\times\omega = 0$, so $V\omega = \omega$. □

3.5 Body perturbations, the scaled frame, and gradients

Lemma 3.5 (First-order body perturbation)**.** *For $x = (t, R)$ and $\xi = (v, \omega)$,*

$$x\exp(\varepsilon\xi^\wedge) = \big(t + \varepsilon Rv + O(\varepsilon^2),\ R + \varepsilon R[\omega]_\times + O(\varepsilon^2)\big). \tag{3.10}$$

For a pure rotation twist ($v = 0$) the translation is unchanged exactly.

Proof. $x\exp(\varepsilon\xi^\wedge) = (t + RV(\varepsilon\omega)\varepsilon v,\ R\exp(\varepsilon[\omega]_\times))$ and $V(0) = I$, $\exp(\varepsilon[\omega]_\times) = I + \varepsilon[\omega]_\times + O(\varepsilon^2)$. □

Definition 3.6 (Body directional derivative)**.** For $f : \mathrm{SE}(3) \to \mathbb{R}$ and a twist ξ,

$$(\mathcal{D}_\xi f)(x) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_{\varepsilon=0} f\big(x\exp(\varepsilon\xi^\wedge)\big). \tag{3.11}$$

This is the derivative of f along the physical motion "move with body twist ξ". It never differentiates the 9 entries of R as independent coordinates (they are not: they satisfy $R^\top R = I$).

Scaled twist frame. Fix speed scales $s_t, s_r > 0$ and set

$$\xi_i = (s_t e_i,\ 0)\ (i = 1, 2, 3), \qquad \xi_{3+j} = (0,\ s_r e_j)\ (j = 1, 2, 3). \tag{3.12}$$

For a function $T(x, y)$ of a pose pair, the *endpoint gradients* are the vectors

$$c_x[i] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_0 T\big(x\exp(\varepsilon\xi_i^\wedge), y\big), \qquad c_y[i] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_0 T\big(x, y\exp(\varepsilon\xi_i^\wedge)\big), \qquad i = 1, \ldots, 6. \tag{3.13}$$

By linearity, if $\xi = \sum_i u_i\xi_i$ then $\mathcal{D}_\xi T = c \cdot u$.

4 Motion model, travel time and the Eikonal equation on SE(3)

4.1 Motion model

Assumption 4.1 (Baseline)**.** A fully actuated rigid body, no obstacles, no acceleration limits, controlled by its body twist $(v(s), \omega(s))$ subject to the *ellipsoidal speed limit*

$$\frac{|v|^2}{s_t^2} + \frac{|\omega|^2}{s_r^2} \le 1. \tag{4.1}$$

Trajectories satisfy $\dot{x} = x\,\xi^\wedge$, i.e.

$$\dot{t} = Rv, \qquad \dot{R} = R[\omega]_\times. \tag{4.2}$$

Writing $\xi = \sum_i u_i \xi_i$, i.e. $v = s_t u_t$, $\omega = s_r u_r$ with $u = (u_t, u_r) \in \mathbb{R}^6$, constraint (4.1) becomes simply $|u| \le 1$.

Definition 4.2 (Metric and travel time)**.** The left-invariant metric on SE(3) is $g_x(\xi,\xi) = |v|^2/s_t^2 + |\omega|^2/s_r^2$ for body twists $\xi = (v, \omega)$; the frame (ξ_i) is g-orthonormal. The length of a trajectory on $[0,\tau]$ is $L = \int_0^\tau \sqrt{g(\xi(s),\xi(s))}\,\mathrm{d}s$. The travel time $T(x,y)$ is the infimum of durations τ of admissible trajectories from x to y.

Since (4.1) says speed ≤ 1, every admissible trajectory has $L \le \tau$, and any trajectory can be reparametrized to unit speed. Therefore

$$T(x,y) = \inf\{L(\gamma) : \gamma \text{ a trajectory from } x \text{ to } y\}, \tag{4.3}$$

the Riemannian distance of g. Note $|\dot{t}| = |Rv| = |v|$, so g is the *product* of the Euclidean metric on $\mathbb{R}^3$ (weight $1/s_t^2$) and the standard metric on SO(3) (weight $1/s_r^2$).

4.2 Symmetries

Proposition 4.3. *For all poses x, y, h: (i) $T(hx, hy) = T(x,y)$ (left invariance), hence $T(x,y) = T(e, x^{-1}y)$; (ii) $T(x,y) = T(y,x)$; (iii) $T(x,y) = 0 \iff x = y$.*

Proof. (i) If $x(s)$ solves $\dot{x} = x\xi^\wedge$ then $hx(s)$ solves $\frac{\mathrm{d}}{\mathrm{d}s}(hx) = (hx)\xi^\wedge$ with the same body twist, hence the same admissibility and duration. (ii) Reverse time: $\tilde{x}(s) = x(\tau - s)$ has body twist $-\xi(\tau - s)$, which satisfies (4.1) because the constraint set is symmetric. (iii) follows from Theorem 5.3 below. □

4.3 Dynamic programming and the Eikonal equation

Proposition 4.4 (Dynamic programming principle)**.** *For $0 \le h \le T(x,y)$,*

$$T(x,y) = \inf_{\text{admissible } \xi(\cdot) \text{ on } [0,h]} \Big[h + T\big(x(h), y\big) \Big]. \tag{4.4}$$

Proof. ($\le$) Following any admissible control for time h and then an (almost) optimal trajectory from $x(h)$ gives a trajectory from x to y of duration $h + T(x(h),y) + \epsilon$. ($\ge$) Restrict an almost optimal trajectory to $[0,h]$. □

Theorem 4.5 (Endpoint Eikonal equations)**.** *At every pair (x,y) where T is differentiable,*

$$|c_x|^2 = 1, \qquad |c_y|^2 = 1. \tag{4.5}$$

Proof. Take a constant normalized control u with $|u| \le 1$ and twist $\xi_u = \sum u_i \xi_i$. By (4.4), $T(x,y) \le h + T(x\exp(h\xi_u^\wedge), y)$. Subtract $T(x,y)$, divide by h and let $h \to 0$: $0 \le 1 + c_x \cdot u$. Taking $u = c_x/|c_x|$ gives $|c_x| \le 1$. Conversely, take an optimal trajectory from x that is continuously differentiable with initial normalized control u^*, $|u^*| = 1$ (Theorem 5.3 provides one). Along it $T(x(h), y) = T(x,y) - h$; differentiating at $h = 0$ gives $c_x \cdot u^* = -1$, and by Cauchy–Schwarz $|c_x| \ge 1$. The statement for c_y follows from symmetry (Proposition 4.3(ii)). □

Remark 4.6 (Hamiltonian form). The proof shows the equation is $\mathcal{H}(c_x) = 1$ with Hamiltonian $\mathcal{H}(c) = \max_{|u|\le 1}(-c \cdot u) = |c|$. In unscaled body coordinates $c = (s_t p_v, s_r p_\omega)$ where p_v, p_ω are the derivatives along unit body translations and rotations, so the equation reads $s_t^2|p_v|^2 + s_r^2|p_\omega|^2 = 1$.

Remark 4.7 (Joint versus endpoint equations). With the product metric $g \oplus g$ on N, the joint squared gradient is $|c_x|^2 + |c_y|^2 = 2$, i.e. the joint equation (2.2) with $S = 1/\sqrt{2}$. The joint equation is *one* scalar condition and is implied by the *two* endpoint conditions, but not conversely: it allows, e.g., $|c_x|^2 = 1.5$, $|c_y|^2 = 0.5$. We therefore use (4.5), as is done in [1].

Remark 4.8 (Other motion models change the Hamiltonian and the decoder together).

Control set	Hamiltonian $\mathcal{H}(p_v, p_\omega)$	Travel time
ellipsoid $\lvert v\rvert^2/s_t^2 + \lvert\omega\rvert^2/s_r^2 \le 1$	$\sqrt{s_t^2\lvert p_v\rvert^2 + s_r^2\lvert p_\omega\rvert^2}$	$\sqrt{\lvert\Delta t\rvert^2/s_t^2 + \theta^2/s_r^2}$
independent $\lvert v\rvert \le s_t,\ \lvert\omega\rvert \le s_r$	$s_t\lvert p_v\rvert + s_r\lvert p_\omega\rvert$	$\max(\lvert\Delta t\rvert/s_t,\ \theta/s_r)$
budget $\lvert v\rvert/s_t + \lvert\omega\rvert/s_r \le 1$	$\max(s_t\lvert p_v\rvert,\ s_r\lvert p_\omega\rvert)$	$\lvert\Delta t\rvert/s_t + \theta/s_r$

The latent Z of Section 6 serves all three; only the decoder changes (Remark 6.7). A decoder must never be paired with the Eikonal loss of a different model. The rest of this report uses the ellipsoidal model.

5 The exact travel time

Lemma 5.1 (Translational lower bound). *Any trajectory from x to y on $[0,\tau]$ satisfies $\int_0^\tau |v|\,\mathrm{d}s \ge |t_y - t_x|$.*

Proof. $|t_y - t_x| = |\int_0^\tau \dot{t}\,\mathrm{d}s| \le \int_0^\tau |Rv|\,\mathrm{d}s = \int_0^\tau |v|\,\mathrm{d}s$. □

Lemma 5.2 (Rotational lower bound). *Any curve $R(s)$, $s \in [0,\tau]$, from R_x to R_y with $\dot{R} = R[\omega]_\times$ satisfies $\int_0^\tau |\omega|\,\mathrm{d}s \ge \theta(R_x^\top R_y)$.*

Proof. Lift to $q(s) \in S^3$ (Q7). By (Q4), the S^3-length of q is $\int|\dot{q}| = \frac{1}{2}\int|\omega|$. By (Q6) this is at least $\arccos(q(0)\cdot q(\tau))$. By (Q5), $q(0)\cdot q(\tau)$ is the scalar part of a quaternion representing $R_x^\top R_y$, so by (Q2) it equals $\pm\cos\frac{\theta}{2}$ with $\theta = \theta(R_x^\top R_y)$. Since $\arccos(\pm\cos\frac{\theta}{2}) \ge \frac{\theta}{2}$ for $\theta \in [0,\pi]$, we get $\frac{1}{2}\int|\omega| \ge \frac{\theta}{2}$. □

Theorem 5.3 (Exact travel time). *Under Assumption 4.1,*

$$\boxed{T(x,y) = \sqrt{\frac{|t_x - t_y|^2}{s_t^2} + \frac{\theta(R_x^\top R_y)^2}{s_r^2}}} \tag{5.1}$$

and an optimal trajectory is, for $\lambda \in [0,1]$ and elapsed time $s = \lambda T$,

$$t(\lambda) = (1-\lambda)t_x + \lambda t_y, \qquad R(\lambda) = R_x \exp\big(\lambda\,\mathrm{Log}(R_x^\top R_y)\big). \tag{5.2}$$

Proof. Lower bound. Let a trajectory have duration τ and put $\alpha = |v|/s_t$, $\beta = |\omega|/s_r$, so $\alpha^2+\beta^2 \le 1$. Using the triangle inequality for integrals of the vector function (α, β),

$$\tau \;\ge\; \int_0^\tau \sqrt{\alpha^2+\beta^2}\,\mathrm{d}s \;\ge\; \sqrt{\Big(\int_0^\tau \alpha\Big)^2 + \Big(\int_0^\tau \beta\Big)^2} \;\ge\; \sqrt{\frac{|\Delta t|^2}{s_t^2} + \frac{\theta^2}{s_r^2}}, \tag{5.3}$$

by Lemmas 5.1 and 5.2. *Attained.* Let $\mathrm{Log}(R_x^\top R_y) = \theta[n]_\times$ (if $\theta = \pi$ pick either axis). Along (5.2) with $s = \lambda T$: $\dot t = \Delta t/T$ with $\Delta t = t_y - t_x$, and $\dot R = R\,[\theta n/T]_\times$, so $|v| = |\Delta t|/T$ and $|\omega| = \theta/T$. The scaled speed is $\sqrt{|\Delta t|^2/s_t^2 + \theta^2/s_r^2}/T = 1$, and the duration is T. □

Remark 5.4 (The optimal path is not a screw motion). A constant body twist generates $t(s) = t_x + R_x V(s\omega)sv$, a helix. The optimal path has *constant world* translational velocity $\Delta t/T$ and constant body angular velocity $\theta n/T$; its body linear velocity $v(s) = R(s)^\top \Delta t/T$ rotates. It coincides with a one-parameter subgroup $x\exp(s\xi^\wedge)$ only when $\Delta t \parallel R_x n$ or $\theta = 0$. This matters for the Bellman operator (Proposition 9.1).

6 The smooth latent representation

6.1 Definition of the latent

Definition 6.1. For $p = (x, y)$ define

$$a(p) = |t_x - t_y|^2, \qquad b(p) = \frac{3 - \mathrm{tr}(R_x^\top R_y)}{4}, \qquad Z(p) = (a(p), b(p)). \tag{6.1}$$

Proposition 6.2 (Three descriptions of the latent)**.** *With $\theta = \theta(R_x^\top R_y)$ and $q = (q_0, q_v)$ a quaternion of $R_x^\top R_y$:*

$$b = \sin^2\frac{\theta}{2} = 1 - q_0^2 = |q_v|^2 = \frac{\|R_x - R_y\|_F^2}{8} \in [0,1], \qquad a = |t_x - t_y|^2. \tag{6.2}$$

In particular Z consists of squared chordal distances *in the embedding $G(x) = (t_x, \mathrm{vec}\,R_x)$: a is the squared distance of the translation blocks and $8b$ of the rotation blocks.*

Proof. By (3.4), $3 - \mathrm{tr} = 2 - 2\cos\theta = 4\sin^2\frac{\theta}{2}$. By (Q3), $3 - \mathrm{tr} = 4 - 4q_0^2$. Finally $\|R_x - R_y\|_F^2 = \mathrm{tr}(2I - R_x^\top R_y - R_y^\top R_x) = 6 - 2\,\mathrm{tr}(R_x^\top R_y)$. □

Corollary 6.3 (Smoothness and invariance)**.** *a and b are polynomials in the entries of $G(x), G(y)$; hence $Z : N \to \mathbb{R}^2$ is smooth (indeed real-analytic) everywhere, including the diagonal $x = y$ and the half-turn set $\theta = \pi$. Both are left-invariant: $Z(hx, hy) = Z(x, y)$, and symmetric: $Z(x, y) = Z(y, x)$.*

6.2 The decoder

Since $b = \sin^2(\theta/2)$ and $\theta \in [0, \pi]$, we have $\theta = 2\arcsin\sqrt{b}$. Define

$$\varphi(b) = 4\arcsin^2\sqrt{b}\ (= \theta^2), \qquad Q(a,b) = \frac{a}{s_t^2} + \frac{\varphi(b)}{s_r^2}, \qquad \boxed{D(a,b) = \sqrt{Q(a,b)}} \tag{6.3}$$

Units: a is length^2, b is dimensionless, Q is time^2. The decoder ignores the G coordinates.

Lemma 6.4 (Properties of φ). *(a) φ is real-analytic on $(-1,1)$ with power series*

$$\varphi(b) = 4\sum_{n\geq 1} c_n b^n = 4b + \frac{4}{3}b^2 + \frac{32}{45}b^3 + \cdots, \qquad c_1 = 1, \quad c_{n+1} = \frac{(2n)^2}{(2n+1)(2n+2)}c_n, \tag{6.4}$$

equivalently $c_n = \dfrac{2^{2n-1}((n-1)!)^2}{(2n)!}$; the radius of convergence is 1.

(b) For $0 < b < 1$, with $\theta = 2\arcsin\sqrt{b}$,

$$\varphi'(b) = \frac{4\arcsin\sqrt{b}}{\sqrt{b(1-b)}} = \frac{4\theta}{\sin\theta}, \qquad \lim_{b\to 0}\varphi'(b) = 4, \qquad \lim_{b\to 1}\varphi'(b) = +\infty. \tag{6.5}$$

(c) (A one-dimensional Eikonal identity) $b(1-b)\,\varphi'(b)^2 = 4\varphi(b)$ for $0 < b < 1$.

Proof. (a) Let $f(u) = \arcsin^2 u$. Then $f' = 2\arcsin u/\sqrt{1-u^2}$, so $\sqrt{1-u^2}f' = 2\arcsin u$; differentiating again gives the ODE $(1-u^2)f'' - uf' = 2$. f is even and analytic on $(-1,1)$; insert $f = \sum_{n\geq 1} c_n u^{2n}$ and compare coefficients: the constant term gives $2c_1 = 2$, and the u^{2n} term gives $(2n+2)(2n+1)c_{n+1} = (2n)^2 c_n$. Since f is a power series in $u^2 = b$, $\varphi(b) = 4f(\sqrt{b}) = 4\sum c_n b^n$ is analytic in b. The ratio $c_{n+1}/c_n \to 1$ gives radius 1. (b) Chain rule; then substitute $\sqrt{b} = \sin\frac{\theta}{2}$, $\sqrt{1-b} = \cos\frac{\theta}{2}$. (c) $b(1-b)\varphi'^2 = 16\arcsin^2\sqrt{b} = 4\varphi$. □

Use the series for small b in code: the closed form of φ' is $0/0$ at $b = 0$ even though φ is perfectly smooth there. The blow-up at $b = 1$ is genuine.

6.3 The factorization theorem

Theorem 6.5 (Smooth graph representation of T on SE(3)). *Let $G(x) = (t_x, \operatorname{vec} R_x) \in \mathbb{R}^{12}$ and*

$$H(x,y) = \big(G(x),\, G(y),\, a(x,y),\, b(x,y)\big) \in \mathbb{R}^{26}. \tag{6.6}$$

Then:

(i) H is a real-analytic embedding of the 12-dimensional manifold $N = \mathrm{SE}(3)\times\mathrm{SE}(3)$; $\Sigma = H(N)$ is a 12-dimensional submanifold of $\mathbb{R}^{26}$.

(ii) $T = D \circ H$ exactly, with D from (6.3).

(iii) $T^2 = Q \circ Z$ is real-analytic on $\{b < 1\}$, i.e. away from half turns, including across the diagonal.

(iv) T is real-analytic on the regular set

$$N_{\mathrm{reg}} = \{(x,y) : x \neq y,\ \theta(R_x^\top R_y) < \pi\}, \tag{6.7}$$

and is not differentiable on its complement (Section 8).

Proof. (i) G is injective, and its differential is injective: by Lemma 3.5 the images of the frame vectors $\xi_1, \ldots, \xi_6$ are $(s_t R_x e_i, 0)$ and $(0, s_r \operatorname{vec}(R_x[e_j]_\times))$, which are linearly independent. Apply Proposition 1.1. (ii) By Proposition 6.2, $\varphi(b) = \theta^2$, then use Theorem 5.3. (iii) φ is analytic on $(-1,1)$ and a, b are polynomials. (iv) $Q > 0$ off the diagonal (it vanishes only when $a = 0$ and $b = 0$) and $\sqrt{\cdot}$ is analytic on $(0,\infty)$. □

Example 6.6 (Consistency with the circle example)**.** For planar rotations $R_x^\top R_y = \exp(\delta[e_3]_\times)$, $b = \sin^2\frac{\delta}{2} = \frac{1-\cos\delta}{2}$. With $z = \cos\delta = 1 - 2b$ one gets $2\arcsin\sqrt{b} = \arccos z$, which is exactly the latent $Z = \cos(x - y)$ and decoder $D = \arccos z$ of the original circle construction. The SE(3) latent is its natural generalization.

Remark 6.7 (Same latent, other decoders). For the models of Remark 4.8 the same Z works with $D_{\max}(a, b) = \max\left(\sqrt{a}/s_t,\, 2\arcsin\sqrt{b}/s_r\right)$ and $D_{\text{sum}}(a, b) = \sqrt{a}/s_t + 2\arcsin\sqrt{b}/s_r$. These decoders have extra non-smoothness (at $a = 0$ and on the switching surface of the max).

7 The Eikonal equation through H on SE(3)

7.1 Block structure of the Jacobian

Let $J_x^L H \in \mathbb{R}^{26\times 6}$ be the Jacobian of H along the six scaled body directions of the start pose, i.e. column i is $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}|_0 H(x\exp(\varepsilon\xi_i^\wedge), y)$. By Lemma 3.5,

$$J_x^L H = \begin{pmatrix} s_t\,[R_x e_1 \;\; R_x e_2 \;\; R_x e_3] & & 0 \\ 0 & & s_r\,[\mathrm{vec}(R_x[e_1]_\times) \;\; \mathrm{vec}(R_x[e_2]_\times) \;\; \mathrm{vec}(R_x[e_3]_\times)] \\ 0_{12\times 3} & & 0_{12\times 3} \\ & J_{Z,x} & \end{pmatrix}, \tag{7.1}$$

where the first two block rows are the 12 coordinates of $G(x)$, the third the 12 coordinates of $G(y)$, and $J_{Z,x} \in \mathbb{R}^{2\times 6}$ is the Jacobian of $Z = (a, b)$ along $\xi_1, \ldots, \xi_6$. Because D depends only on (a, b), $\nabla D = (0_{24},\; \partial_a D,\; \partial_b D)$, and the chain rule collapses to

$$c_x = (J_x^L H)^\top \nabla D = J_{Z,x}^\top \nabla_Z D, \qquad \nabla_Z D = \Big(\frac{1}{2s_t^2 T},\; \frac{\varphi'(b)}{2s_r^2 T}\Big). \tag{7.2}$$

Similarly $c_y = J_{Z,y}^\top \nabla_Z D$. The endpoint Eikonal equations become

$$\nabla_Z D^\top K_x \nabla_Z D = 1, \qquad \nabla_Z D^\top K_y \nabla_Z D = 1, \qquad K_x = J_{Z,x} J_{Z,x}^\top,\;\; K_y = J_{Z,y} J_{Z,y}^\top \in \mathbb{R}^{2\times 2}. \tag{7.3}$$

This form holds for *any* differentiable latent (learned or exact); only the 2×2 Gram matrices K_x, K_y of the latent enter.

7.2 Derivatives of the exact latent

Lemma 7.1. *On N_{reg} write $\Delta t = t_y - t_x$, $R_x^\top R_y = \exp(\theta[n]_\times)$, $0 < \theta < \pi$. Then the gradients of a, b in the scaled frame at the start pose are*

$$\nabla_x a = \begin{pmatrix} -2s_t\, R_x^\top \Delta t \\ 0 \end{pmatrix}, \qquad \nabla_x b = \begin{pmatrix} 0 \\ -\frac{s_r}{2}\sin\theta\; n \end{pmatrix}, \tag{7.4}$$

and at the goal pose

$$\nabla_y a = \begin{pmatrix} 2s_t\, R_y^\top \Delta t \\ 0 \end{pmatrix}, \qquad \nabla_y b = \begin{pmatrix} 0 \\ \frac{s_r}{2}\sin\theta\; n \end{pmatrix}. \tag{7.5}$$

Proof. Translation directions at x. By Lemma 3.5, $t_x \mapsto t_x + \varepsilon s_t R_x e_i$, R_x unchanged to first order, so $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}|t_x + \varepsilon s_t R_x e_i - t_y|^2 = 2s_t(t_x - t_y)\cdot R_x e_i = -2s_t(R_x^\top \Delta t)_i$, and b is unchanged. *Rotation directions at x.* With $\omega = s_r e_j$, t_x is unchanged and $(R_x\exp(\varepsilon[\omega]_\times))^\top R_y = \exp(-\varepsilon[\omega]_\times)M$, $M = R_x^\top R_y$. Then $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\operatorname{tr}(\exp(-\varepsilon[\omega]_\times)M) = -\operatorname{tr}([\omega]_\times M) = \omega\cdot(M - M^\top)^\vee$ by Lemma 3.1(e), and $(M - M^\top)^\vee = 2\sin\theta\, n$ by (3.4). Hence $\frac{\mathrm{d}b}{\mathrm{d}\varepsilon} = -\frac{1}{4}\cdot 2\sin\theta\,(\omega\cdot n) = -\frac{s_r}{2}\sin\theta\, n_j$. *At y.* Translation: $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}|t_y + \varepsilon s_t R_y e_i - t_x|^2 = 2s_t(R_y^\top \Delta t)_i$. Rotation: $\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\operatorname{tr}(M\exp(\varepsilon[\omega]_\times)) = \operatorname{tr}(M[\omega]_\times) = \operatorname{tr}([\omega]_\times M) = -\omega\cdot 2\sin\theta\, n$, so $\frac{\mathrm{d}b}{\mathrm{d}\varepsilon} = +\frac{s_r}{2}\sin\theta\, n_j$. □

Remark 7.2 (Frames of the axis). n is the axis of $M = R_x^\top R_y$, expressed in the body frame of x. Because $Mn = n$, $R_x n = R_y n$: the world axis is the same seen from either endpoint, and n is also its coordinate vector in the body frame of y.

7.3 Verification of the endpoint Eikonal equations

Theorem 7.3. *On* N_{reg},

$$c_x = \frac{1}{T}\begin{pmatrix} -R_x^\top \Delta t/s_t \\ -\theta\, n/s_r \end{pmatrix}, \qquad c_y = \frac{1}{T}\begin{pmatrix} R_y^\top \Delta t/s_t \\ \theta\, n/s_r \end{pmatrix}, \tag{7.6}$$

and $|c_x|^2 = |c_y|^2 = \dfrac{|\Delta t|^2/s_t^2 + \theta^2/s_r^2}{T^2} = 1.$

Proof. Combine (7.2) with Lemma 7.1. Translation part: $\frac{1}{2s_t^2 T}(-2s_t R_x^\top \Delta t) = -R_x^\top \Delta t/(s_t T)$. Rotation part: $\frac{\varphi'(b)}{2s_r^2 T}\big(-\frac{s_r}{2}\sin\theta\, n\big) = \frac{4\theta/\sin\theta}{2s_r^2 T}\cdot\big(-\frac{s_r}{2}\sin\theta\big)n = -\frac{\theta n}{s_r T}$, using Lemma 6.4(b). The $\sin\theta$ produced by $\mathrm{d}b$ cancels the $1/\sin\theta$ in φ'. The y case is identical with signs flipped. □

Remark 7.4. This is a direct check of Theorem 4.5 through the graph parametrization, and the formulas in (7.6) were also confirmed numerically by central finite differences on random pose pairs (errors $\sim 10^{-9}$).

7.4 The latent metric: the decoder is a distance function

Proposition 7.5 (Latent Gram matrix)**.** *For the exact latent, on* N_{reg},

$$K_x = K_y = \begin{pmatrix} 4s_t^2\, a & 0 \\ 0 & s_r^2\, b(1-b) \end{pmatrix}. \tag{7.7}$$

In particular K *depends only on* Z *itself, and the endpoint Eikonal equations* (7.3) *become a PDE on the two-dimensional latent domain* $\Omega = [0,\infty)\times[0,1]$*:*

$$4s_t^2\, a\,(\partial_a D)^2 + s_r^2\, b(1-b)\,(\partial_b D)^2 = 1, \qquad D(0,0) = 0. \tag{7.8}$$

Proof. From Lemma 7.1, $|\nabla_x a|^2 = 4s_t^2 a$, $|\nabla_x b|^2 = \frac{s_r^2}{4}\sin^2\theta = s_r^2 \sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2} = s_r^2 b(1-b)$, and $\nabla_x a \cdot \nabla_x b = 0$ (disjoint blocks). Same at y. That D solves (7.8): $4s_t^2 a\frac{1}{4s_t^4 T^2} + s_r^2 b(1-b)\frac{\varphi'^2}{4s_r^4 T^2} = \frac{a/s_t^2 + \varphi/s_r^2}{T^2} = 1$ by Lemma 6.4(c). □

Theorem 7.6 (Flat latent coordinates)**.** *Equip the interior of* Ω *with the metric whose inverse is* K*:*

$$\hat{g} = \frac{\mathrm{d}a^2}{4s_t^2 a} + \frac{\mathrm{d}b^2}{s_r^2\, b(1-b)}. \tag{7.9}$$

In the coordinates $r = \sqrt{a} \in [0,\infty)$, $\theta = 2\arcsin\sqrt{b} \in [0,\pi]$ *this metric is flat,* $\hat{g} = \mathrm{d}r^2/s_t^2 + \mathrm{d}\theta^2/s_r^2$*, and* D *is the* $\hat{g}$*-distance from the origin:* $D = \sqrt{r^2/s_t^2 + \theta^2/s_r^2}$*. Moreover, for fixed* y*, the map* $x \mapsto Z(x,y)$ *is a Riemannian submersion from* $(\mathrm{SE}(3), g)$ *onto* $(\Omega^\circ, \hat{g})$ *on the region where* $a > 0$*,* $0 < b < 1$*.*

Proof. $a = r^2 \Rightarrow \mathrm{d}a = 2r\,\mathrm{d}r \Rightarrow \mathrm{d}a^2/(4s_t^2 a) = \mathrm{d}r^2/s_t^2$. $b = \sin^2\frac{\theta}{2} \Rightarrow \mathrm{d}b = \frac{1}{2}\sin\theta\,\mathrm{d}\theta \Rightarrow \mathrm{d}b^2/(s_r^2 b(1-b)) = \frac{\frac{1}{4}\sin^2\theta\,\mathrm{d}\theta^2}{s_r^2\frac{1}{4}\sin^2\theta} = \mathrm{d}\theta^2/s_r^2$. The (r,θ) domain is the convex quadrant strip $[0,\infty)\times[0,\pi]$ with a scaled Euclidean metric, so distances from the origin are scaled Euclidean norms. The submersion statement is $J_{Z,x}J_{Z,x}^\top = \hat{g}^{-1}(Z)$, which is Proposition 7.5. □

Interpretation. The motion metric on the 12-dimensional pair space reduces, through the smooth invariants (a, b), to a 2-dimensional latent space. In the *non-smooth* coordinates $(r, \theta) = (|\Delta t|, \theta)$ the problem is trivially flat; the smooth coordinates $(a, b) = (r^2, \sin^2 \frac{\theta}{2})$ are the "folded" versions of (r, θ), even under $r \mapsto -r$ and under $\theta \mapsto 2\pi - \theta$. The folding is precisely what makes Z smooth, and unfolding is precisely what D does.

7.5 Gradient flow reproduces the optimal path

Proposition 7.7. *On N_{reg}, the steepest-descent control $u = -c_x/|c_x| = -c_x$ gives body twist $\xi = \big(R_x^\top \Delta t/T,\ \theta n/T\big)$, which is exactly the initial twist of the optimal path (5.2). Along the continuous flow $\dot{x} = x\,\xi(x)^\wedge$, $\frac{\mathrm{d}}{\mathrm{d}s} T(x(s), y) = c_x \cdot u = -1$, the translation moves on the straight segment to t_y and the rotation follows the geodesic, so the flow reaches y in time $T(x_0, y)$.*

Proof. $\xi = \sum_i u_i \xi_i = (s_t u_t, s_r u_r)$ with $u = -c_x$ from (7.6). At every point the world translational velocity is $R\,R^\top \Delta t/T = \Delta t/T$, pointing at t_y, and the relative rotation about the fixed world axis decreases at constant rate, which is (5.2). □

Remark 7.8 (Ambient induced metric on SE(3)). From the block form of $J_x^L H$ and Lemma 3.1(b), $\langle \mathrm{vec}(R_x[e_j]_\times), \mathrm{vec}(R_x[e_k]_\times)\rangle = 2\delta_{jk}$, hence

$$(J_x^L H)^\top J_x^L H = \mathrm{diag}(s_t^2 I_3,\ 2s_r^2 I_3) + J_{Z,x}^\top J_{Z,x} \neq I_6. \tag{7.10}$$

So H is not isometric for g (even the G part alone has the factor 2 on rotations), and using the Euclidean geometry of $\Sigma \subset \mathbb{R}^{26}$ in place of the transported metric would give a wrong Eikonal equation.

8 Singularities

8.1 The diagonal $x = y$

Near $y = x$, $T(x, x\exp(\varepsilon\xi^\wedge)) = |\varepsilon|\sqrt{g(\xi, \xi)} + O(\varepsilon^2)$: a cone. T is not differentiable on the diagonal, and c_x has unit length but discontinuous direction there. In latent terms $Z \to (0, 0)$ smoothly, Q is analytic, and the cusp comes from $\sqrt{Q}$. Value constraints ($T(x, x) = 0$) are imposed instead of derivative constraints; it is convenient to impose them through Q, avoiding the derivative of $\sqrt{\cdot}$ at 0.

8.2 The rotational cut locus $\theta = \pi$

Proposition 8.1. *T is not differentiable at any pair with $\theta(R_x^\top R_y) = \pi$. Along the curve $R_y(s) = R_x \exp((\pi + s)[n]_\times)$, t_y fixed, the angle is $\theta(s) = \pi - |s|$ and T has a concave corner (a local maximum with a kink). Meanwhile $b(s) = \sin^2 \frac{\pi+s}{2} = \cos^2 \frac{s}{2}$ is smooth, with a smooth maximum $b = 1$ at $s = 0$.*

Proof. The angle of $\exp(\phi[n]_\times)$ is ϕ for $\phi \in [0, \pi]$ and $2\pi - \phi$ for $\phi \in [\pi, 2\pi]$. Substituting into (5.1) gives a function of $|s|$; since $\partial T/\partial\theta = \theta/(s_r^2 T)$ and $\theta'(s) = \mp 1$ for $s \gtrless 0$, the one-sided derivatives at $s = 0^\pm$ are $\mp\pi/(s_r^2 T) \neq 0$, so T has a corner. That $b(s)$ is smooth follows from $\sin^2 \frac{\pi+s}{2} = \cos^2 \frac{s}{2}$. □

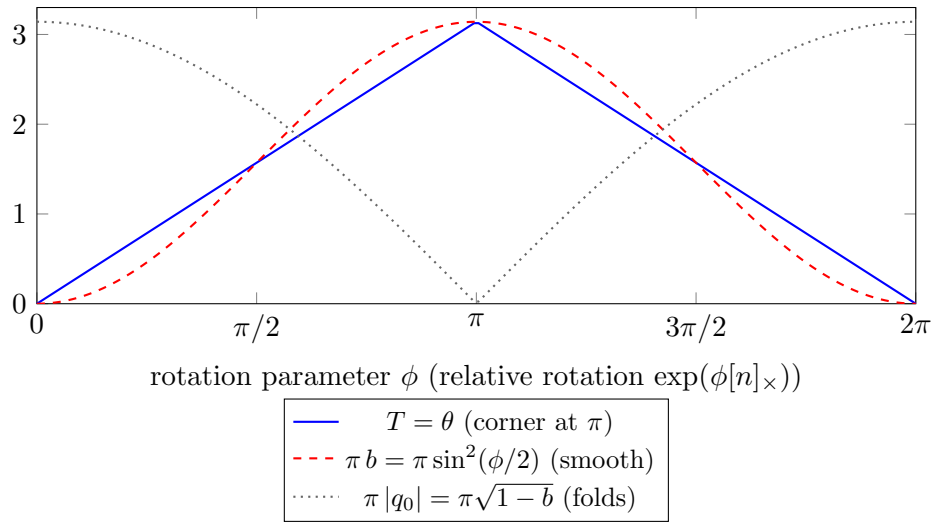


Figure 1: Pure rotation slice ($\Delta t = 0$, $s_r = 1$). The latent b is smooth through the half turn; the decoder $2\arcsin\sqrt{b} = 2\arccos\sqrt{1-b} = 2\arccos|q_0|$ re-creates the corner through the absolute value $|q_0| = \sqrt{1-b}$.

Mechanism. By (Q2) and (Q1), the two quaternions $\pm q$ of the relative rotation give two candidate half-angles $\arccos q_0$ and $\arccos(-q_0)$; the true one is the smaller, $\arccos|q_0|$. Locally T is the *minimum of two smooth branches*, which is a concave corner. Since $b = 1 - q_0^2$ is even in q_0, it is smooth, and the decoder contains $\sqrt{1-b} = |q_0|$. In the chain rule, $\varphi'(b) \to \infty$ as $b \to 1$ while $\nabla_x b \propto \sin\theta \to 0$; for the exact latent the product stays bounded (Theorem 7.3) but its direction $\mp n$ jumps. For a *learned* b_θ near 1 nothing forces ∇b_θ to vanish at the matching rate, so the Eikonal residual is numerically unstable in a band around the cut locus.

Size of the singular set. The half turns form a 2-dimensional subset ($\cong \mathbb{RP}^2$) of the 3-dimensional SO(3), so the cut locus has codimension 1 in N; the diagonal has codimension 6. Both have measure zero, so N_{reg} has full measure. However, under Haar-uniform sampling of rotations the relative angle has density $(1-\cos\theta)/\pi$ on $[0,\pi]$, which is *largest* at $\theta = \pi$, so a substantial fraction of samples lies close to the cut locus.

8.3 Why the Eikonal equation alone does not select T

Example 8.2 (Spurious almost-everywhere solutions)**.** On the circle, with relative angle $\delta \in [-\pi, \pi]$ and unit speed, the true travel time from the goal is $T(\delta) = |\delta|$: it satisfies $|T'| = 1$ off $\{0, \pm\pi\}$, $T(0) = 0$, and has a concave corner at the cut point $\pm\pi$. The function

$$\tilde{T}(\delta) = \begin{cases} |\delta|, & |\delta| \le \pi/2, \\ \pi - |\delta|, & \pi/2 \le |\delta| \le \pi, \end{cases}$$

is continuous on the circle, satisfies $\tilde{T}(0) = 0$ and $|\tilde{T}'| = 1$ almost everywhere as well, yet it is wrong: it has a spurious *convex* corner at $\pm\pi$ where it drops to 0 (a fake second goal). Any function built from unit-slope pieces satisfies the equation off its kinks.

The selection principle is the *viscosity* notion: a solution may have concave corners (minimum of branches, like the cut locus) but no convex corners (maximum of branches) except at the boundary/source. Distance functions are exactly the viscosity solutions. A sampled, almost-everywhere

Eikonal loss does not see corners, so one adds value-based constraints that encode the dynamic programming principle (Section 9).

8.4 The squared-time residual

Since $Q = T^2$ is analytic across the diagonal, one may consider $|c_{Q,x}|^2 = 4Q$ where $c_{Q,x}$ is the gradient of Q (from $c_Q = 2Tc$). This avoids the $\sqrt{\cdot}$ derivative at $x = y$ but not the half-turn singularity in φ. It admits the spurious solution $Q \equiv 0$ after clearing factors, so it must never be used without calibration, boundary and Bellman information and a collapse test.

9 Learning the graph

9.1 Model

Keep G fixed and learn only the latent:

$$H_\vartheta(x,y) = \big(G(x), G(y), Z_\vartheta(x,y)\big), \qquad Z_\vartheta = (a_\vartheta, b_\vartheta), \qquad T_\vartheta = D \circ Z_\vartheta, \tag{9.1}$$

with D the known decoder (6.3). By Proposition 1.1, H_ϑ is an embedding for every smooth Z_ϑ; the architecture is left open. Optional structural choices:

- *Symmetry by exchange averaging:* $\tilde{Z}(x,y) = \frac{1}{2}\big(\mathrm{raw}(x,y) + \mathrm{raw}(y,x)\big)$.
- *Range adapter:* $a_\vartheta = \mathrm{softplus}(\tilde{Z}_1) > 0$, $b_\vartheta = \sigma(\tilde{Z}_2) \in (0,1)$. This keeps D defined but cannot reach $a = 0$ or $b \in \{0,1\}$ exactly, so goal values and half-turn values are only approximated.
- *Left invariance (obstacle-free only):* feeding the network only the relative pose $x^{-1}y$ builds in $Z(hx, hy) = Z(x,y)$ and reduces the input to a function on SE(3). This symmetry is destroyed by obstacles.
- *Rotation input:* use rotation matrices; quaternion inputs must give identical outputs for q and $-q$.

9.2 Losses

Let μ_{reg} be a sampling distribution on N_{reg} (with numerical bands excluded around the diagonal and the cut locus, chosen from the *input* geometry, not from the predicted b_ϑ, so the model cannot evade the loss).

Endpoint Eikonal loss.

$$c_x^\vartheta = J_{Z_\vartheta,x}^\top \nabla_Z D(Z_\vartheta), \quad c_y^\vartheta = J_{Z_\vartheta,y}^\top \nabla_Z D(Z_\vartheta), \qquad \mathcal{L}_E(\vartheta) = \mathbb{E}_{\mu_{\mathrm{reg}}}\Big[\big(|c_x^\vartheta|^2 - 1\big)^2 + \big(|c_y^\vartheta|^2 - 1\big)^2\Big]. \tag{9.2}$$

In practice $c_x^\vartheta[i] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}|_0 T_\vartheta(x\exp(\varepsilon\xi_i^\wedge), y)$ is obtained by differentiating the scalar composite with respect to a zero local twist; no full Jacobian is formed. Training requires second derivatives (derivatives of c^ϑ with respect to ϑ).

Goal (boundary) loss. With a reference length L_{ref} and $\tau_{\mathrm{ref}} = L_{\mathrm{ref}}/s_t$,

$$\mathcal{L}_{\mathrm{goal}} = \mathbb{E}_x\big[Q(Z_\vartheta(x,x))\big]/\tau_{\mathrm{ref}}^2, \tag{9.3}$$

using $Q = T^2$ to avoid differentiating $\sqrt{\cdot}$ at 0.

Bellman (dynamic-programming) loss. For a step $h > 0$ and the normalized control ball $\mathcal{U} = \{|u| \le 1\}$ define the constant-twist Bellman operator

$$(\mathcal{B}_h T)(x, y) = \min_{u \in \mathcal{U}} \Big[h + T\big(x \exp(h\, \xi_u^\wedge), y\big) \Big], \qquad \xi_u = \textstyle\sum_i u_i \xi_i, \tag{9.4}$$

modified near the goal to charge only the elapsed time to arrival when y is reachable within the step. The loss is

$$\mathcal{L}_B = \mathbb{E}\big[(T_\vartheta(x, y) - \mathrm{sg}[(\mathcal{B}_h \bar{T})(x, y)])^2\big] / \tau_{\mathrm{ref}}^2, \tag{9.5}$$

with $\bar{T}$ a slowly updated (e.g. exponential-moving-average) target and sg meaning no gradient flows through the target. In practice the minimum is over a finite sample of controls, which only approximates $\mathcal{B}_h$.

Proposition 9.1 (The exact T is an approximate fixed point of $\mathcal{B}_h$)**.** *For the exact T, $0 < h \le T(x, y)$ and $h s_r < 2\pi$,*

$$0 \;\le\; (\mathcal{B}_h T)(x, y) - T(x, y) \;\le\; h\Big(1 - \frac{\sin(h s_r/2)}{h s_r/2}\Big) = \frac{s_r^2 h^3}{24} + O(h^5). \tag{9.6}$$

Proof. Lower bound: by Proposition 4.4 every term in the minimum is $\ge T(x, y)$. *Upper bound:* let $x^*(\tau)$ be the point at time τ on the optimal path (5.2); then $T(x^*(\tau), y) = T - \tau$. The relative pose $x^{-1} x^*(\tau) = \big(\frac{\tau}{T} R_x^\top \Delta t,\ \exp(\frac{\tau\theta}{T}[n]_\times)\big)$ equals $\exp(\zeta^\wedge)$ for the twist $\zeta = (V(\omega_\zeta)^{-1} \frac{\tau}{T} R_x^\top \Delta t,\ \omega_\zeta)$, $\omega_\zeta = \frac{\tau\theta}{T} n$. By Lemma 3.4 with $\kappa = \kappa(\tau\theta/T)$,

$$g(\zeta, \zeta) \le \kappa^2 \frac{\tau^2 |\Delta t|^2}{T^2 s_t^2} + \frac{\tau^2 \theta^2}{T^2 s_r^2} \le \kappa^2 \tau^2.$$

Choose $\tau = h/\kappa(h\theta/T) \le h$. Since κ is increasing, $\sqrt{g(\zeta, \zeta)} \le \kappa(h\theta/T)\tau = h$, so the constant twist ζ/h is admissible for duration h and lands on $x^*(\tau)$. Thus $\mathcal{B}_h T \le h + T - \tau = T + h(1 - 1/\kappa(h\theta/T))$. Finally $\theta/T \le s_r$ by (5.1), and $1 - 1/\kappa(\phi) = 1 - \frac{\sin(\phi/2)}{\phi/2} = \phi^2/24 + O(\phi^4)$. □

The defect is caused entirely by Remark 5.4: a constant twist moves on a helix, the optimal path on a straight line. It vanishes as $h \to 0$ (numerically: halving h divides it by ≈ 8). Over the $\approx T/h$ steps of a rollout the accumulated bias is $O(h^2)$.

Calibration and supervision (diagnostics). With oracle labels $T^\star$, $Z^\star$:

$$\mathcal{L}_T = \mathbb{E}\big[(T_\vartheta - T^\star)^2\big] / \tau_{\mathrm{ref}}^2, \qquad \mathcal{L}_Z = \mathbb{E}\Big[\frac{(a_\vartheta - a^\star)^2}{L_{\mathrm{ref}}^4} + (b_\vartheta - b^\star)^2\Big]. \tag{9.7}$$

A physics-informed objective is, e.g., $\mathcal{L} = \mathcal{L}_E + \mathcal{L}_B + \lambda_{\mathrm{goal}} \mathcal{L}_{\mathrm{goal}} + \lambda_{\mathrm{anchor}} \mathcal{L}_T$, where $\mathcal{L}_T$ is restricted to a declared anchor set (or $\lambda_{\mathrm{anchor}} = 0$).

Triangle inequality (optional). T satisfies $T(x, y) \le T(x, z) + T(z, y)$, but neither the decoder nor the Eikonal residual enforces it. One may add $\mathbb{E}[\max(0, T_\vartheta(x, y) - T_\vartheta(x, z) - T_\vartheta(z, y))^2]$.

9.3 What the losses can and cannot identify

Proposition 9.2 (Non-identifiability of the latent)**.** *The losses $\mathcal{L}_E, \mathcal{L}_B, \mathcal{L}_{\text{goal}}, \mathcal{L}_T$ depend on Z_ϑ only through $T_\vartheta = D \circ Z_\vartheta$ (and its derivatives). Any two latents with the same $D \circ Z$ give identical losses.*

Example 9.3. $Z' = (s_t^2 T^2,\, 0)$ decodes exactly to T because $D(s_t^2 T^2, 0) = T$. But $T^2 = a/s_t^2 + \theta^2/s_r^2$ is not smooth at the cut locus (since θ has a corner there), so Z' is not smooth. Hence among exact representations, *smoothness* is what singles out latents like (a, b) whose corner is created by the decoder. A smooth network cannot represent Z' exactly and would have to round its corner; whether gradient training prefers the intended smooth latent is an empirical question. Physics-only outputs $(a_\vartheta, b_\vartheta)$ must not be read as "translation" and "rotation" components without evidence; $\mathcal{L}_Z$ tests that interpretation.

10 Path extraction

Gradient rollout. On regular pairs take

$$u_k = -c_x^\vartheta / |c_x^\vartheta|, \qquad \xi_k = (s_t u_{k,t},\, s_r u_{k,r}), \qquad x_{k+1} = x_k \exp(h\, \xi_k^\wedge). \tag{10.1}$$

For the exact T this is a time discretization of the flow of Proposition 7.7. The gradient must be that of the *decoded* time T_ϑ, not of a latent coordinate. Near the cut locus or the goal the direction is unreliable; a sampling step is used instead:

$$u_k = \arg\min_{u \in \mathcal{U}_K} \left[h + T_\vartheta(x_k \exp(h \xi_u^\wedge), y) \right] \tag{10.2}$$

over a finite candidate set $\mathcal{U}_K$ of normalized controls.

Path cost. The cost of a trajectory is $\int \sqrt{|v|^2/s_t^2 + |\omega|^2/s_r^2}\, \mathrm{d}s$ (the g-length), never an unweighted norm of pose coordinates. The ratio to $T(x_0, y)$ measures sub-optimality.

11 Summary: what is proved and what is open

Proved here (obstacle-free, ellipsoidal speed limit).

1. The exact travel time is $T = \sqrt{|\Delta t|^2/s_t^2 + \theta^2/s_r^2}$, the distance of a product metric (Theorem 5.3).
2. $T = D \circ H$ with $H = (G(x), G(y), a, b)$ a real-analytic embedding of N into $\mathbb{R}^{26}$ and $Z = (a, b)$ the squared chordal distances in the embedding; all non-smoothness of T lives in the explicit decoder $D = \sqrt{a/s_t^2 + 4 \arcsin^2 \sqrt{b}/s_r^2}$ (Theorem 6.5).
3. Both endpoint Eikonal equations hold through H; through the chain rule they reduce to a 2×2 Gram condition on the latent (equations (7.3), Theorem 7.3).
4. The latent carries a metric $\hat{g}$ that is flat in folded coordinates, and D is the $\hat{g}$-distance from the origin (Theorem 7.6).
5. The singular set is the diagonal (cusp) and the half-turn cut locus (concave corner, min of two quaternion branches); the latent is smooth across both (Section 8).
6. The exact T is a fixed point of the constant-twist Bellman operator up to $O(h^3)$ per step (Proposition 9.1).

Not established (research hypotheses).

1. That a neural network trained with these losses converges to the viscosity solution, or trains more reliably near cut loci than a network that outputs T directly.

2. That obstacle environments admit a globally smooth, finite-dimensional latent with a tractable decoder. With obstacles, left invariance is lost, a and b lose their physical meaning, and new singularities (cut loci created by obstacles) appear; richer latents or min-type decoders [4] are natural candidates.

3. Novelty relative to transformed distance fields [2], Riemannian distance-field learning [3] and latent metric models [1].

The research question. Find a smooth, configuration-dependent latent Z and a tractable decoder D such that (i) the correct travel time is representable as $D \circ Z$, (ii) the transformed derivative loss is stable on the regular set, and (iii) training selects the intended global solution. The SE(3) construction above proves feasibility in the obstacle-free case and makes the mechanism explicit; it does not by itself guarantee successful learning.

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